

Q1) a) • Merge sort

b) • guaranteed Performance

Merge sort provides a stable and guaranteed time complexity of $O(n \log n)$ across all cases (best, average and worst)

- Reliability : for processing critical official data like 50,000 student marks, Quick sort runs like a degenerating to $O(n^2)$ if worst case conditions are met, whereas Merge sort remains consistently fast.

- Stability : It is a stable sorting algorithm meaning students with identical marks will retain their original relative order

c) • Time Complexity : $O(n \log n)$

- Auxiliary Space Complexity : $O(n)$ because it requires temporary extra array's to merge the divided sub-array's.

d) One Limitation.

- High Memory Overhead : It is not an in place sorting algorithm it requires extra memory proportional to the size of the input data ($O(n)$ auxiliary space) which can be inefficient under extreme memory constraints.

Q27 Ans : Algorithm : Quick Sort
(Divide and Conquer)

- Pivot Impact

Balanced (Good Pivot) : equal splits minimize recursion depth, giving optimal performance.

- unbalanced (Poor Pivot)

Extreme split (0 and $n-1$) destroys divide and Conquer efficiency, causing a performing collapse.

→ Complexities

1) Best Case Time : $O(n \log n)$

2) Worst Case : $O(n^2)$

3) Auxiliary Space : $O(\log n)$

* Improvement

Use Randomize Pivot or Median of Three to ensure Balanced partitions and prevent $O(n^2)$ worst case Behavior.



Vidyavardhini's College of Engineering and Technology

Department of Artificial Intelligence & Data Science

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Application Assignment – Divide and Conquer Approach

Course Code/Name: 2483113/ Analysis of Algorithm

Year/Sem: S.E./ III

Name of Student:	Shaurya Jamsundekar		
Roll No. :	22	Division:	A
Application Assignment Title:	Divide & Conquer		
Date of Submission:		Date of correction:	

Evaluation

Performance Indicator		Max. Marks	Marks Obtained
Algorithm Selection		2	
Algorithm Application		3	
Divide and Conquer Process		3	
Complexity Analysis		2	
Total		10	
Performance Indicator	Exceed Expectations (EE)	Meet Expectations (ME)	Below Expectations (BE)
Algorithm Selection (2 Marks)	Select the most appropriate Divide and Conquer algorithm with complete justification. (2 Marks)	Appropriate algorithm selected with minor justification gaps. (1 Mark)	Incorrect algorithm selected or no justification provided. (0 Marks)
Algorithm Application (3 Marks)	Correctly applies the algorithm and obtains the expected result (3 Marks)	Applies the algorithm with minor logical or procedural errors. (2 Marks)	Unable to apply the algorithm correctly or produce incorrect results. (0-1 Marks)
Divide and Conquer Process (3 Marks)	Clearly demonstrates all Divide, Conquer, and Combine steps (3 Marks)	Demonstrates most phases with minor omissions or inaccuracies. (2 Marks)	Inappropriate techniques or weak/no justification. (0-1 Marks)
Complexity Analysis (2 Marks)	Correctly determines and explains time and auxiliary space complexity. (2 Marks)	Correctly identifies complexities but provides limited or incomplete justification. (1 Mark)	Incorrect complexity analysis (0 Marks)

Checked by

Name of Faculty : Prof. Sneha Yadav.

Signature :

Date :

